

برنامه ریزی و طراحی فرودگاه
(مسائل سری 6)

(1)

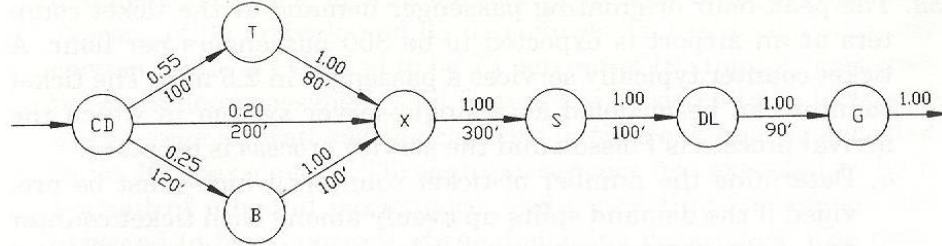


Figure B-10 Link-and-node diagram for enplaning system.

TABLE B-21 Characteristics of Passenger-Processing System

Component	Number of servers	Service rate per person, s
Curb-front doors (CD)	3	5
Ticket counters (T)	6	120
Baggage check-in (B)	2	50
Security (X)	2	15
Seat selection (S)	4	30
Boarding devices (G)	3	6

35. An originating passenger-processing network is to be analyzed during the peak hour. The peak-hour demand is 300 passengers. The passengers proceed from the curb front, to the ticketing and baggage check-in area, to security, to seat selection, to the departure lounge, and then to the boarding device. The distribution of passengers and the walking distance in ft between these processors are shown in Fig. B-10. There is an average of 0.25 visitor per passenger, and 40 percent of visitors proceed beyond security. Average walking speeds are 2 ft/s. The characteristics of the passenger-processing components are given in Table B-21. The system exhibits Poisson arrival and constant service distributions, and the passenger demand is split equally between each processor of a particular type. Delays at the departure lounge are not considered in this analysis. Assume that the period during which demand exceeds capacity is limited to 10 min.

- Draw a network which shows the number of persons per hour who proceed between processing components.
- Determine the average passenger-processing time during the peak hour.
- Determine the number of ticketing processing positions and the number of baggage check-in positions required if the average line length at each is not to exceed 5 passengers in the peak hour.

2) یک پایانه دارای 10 محل توقف هواپیما (Gate) برای هواپیماهای تا 500 صندلی و 30 محل توقف برای هواپیماها با 200 صندلی است. متوسط ضریب اشغال (Load Factor) برابر 60 درصد برای همه هواپیماهاست. زمان اشغال توقفگاه $(Gate \text{ (Gate) Time}) = (0.6 + 0.0005S)$ ساعت بر هواپیما، وقتی صندلی بر هواپیما $S =$ است. 75 درصد از مسافرینی که با هواپیما به فرودگاه می‌رسند فرودگاه را ترک می‌کنند و جای خود را با مسافران خروجی می‌دهند. پیدا کنید:

الف) ماکزیمم ظرفیت مسافر پایانه

ب) ظرفیت پایانه اگر 15% از هواپیماها دارای 400 صندلی، 35% دارای 180 صندلی و 50% دارای 100 صندلی باشد.

3)

34. The arrival distribution of passengers at the initial originating passenger-processing positions during a 40-min design period is given in Table B-19. These positions consist of ticketing, baggage check-in, and gate information positions. Performance and use data indicating the type of passenger processing required at these positions and the average service time are given in Table B-20. Five passenger-processing positions exist, and each is capable of providing all the above services. Assume that the arrival distribution is Poisson and that the service rate is constant.
- Find the average passenger service time at the initial passenger-processing position.
 - Find the average delay to passengers during the design period.
 - Find the average queue length during the design period.

TABLE B-19 Passenger Arrival Distribution for Design Period

Period	Passengers arriving in period	Period	Passengers arriving in period
9:00–9:05	4	9:20–9:25	28
9:05–9:10	8	9:25–9:30	19
9:10–9:15	14	9:30–9:35	8
9:15–9:20	26	9:35–9:40	2

TABLE B-20 Facility Passenger Servicing Data

Type of service required	Passengers requiring service, %	Average service time, s
Ticket only	20	80
Baggage only	15	60
Ticket and baggage	45	150
Information only	20	30